EE5203 L01 Practice Worksheet

From C code to a blinking LED: what each step proves - Basri Erdogan

Time: 30-45 minutes

Rule: Try every question before you open the solutions. Short answers are fine. One sentence is usually enough.

Part A - Name the parts (5 minutes)

1. Write one word from this list next to each item: **processor**, **microcontroller**, **peripheral**, **board**, **embedded system**.

Item	Your word
Arm Cortex-M4	
STM32F401RE	
Nucleo-F401RE	
TIM2 (a timer inside the chip)	
A thermostat: STM32 board + temperature sensor + relay	

2. An embedded system has limits. Give one example for each limit. Example for time: *"it must react within 10 ms"*.
   a. Memory: …
   b. Energy: …
   c. Safety: …
3. The console says **"Build: 0 errors"**, but the LED does not blink. Is this possible? **Yes / No.** Explain in one sentence.

Part B - The path from code to LED (10 minutes)

4. Put the six steps in order. Write 1 (first) to 6 (last).

Step	Order
Flash the firmware into the chip	
Link the object files into one program	
Edit `main.c`	
Compile the C files	
Watch the LED	
Look at a variable at a breakpoint	

5. Each message comes from one step of the path. Match them. Steps: **compile**, **link**, **flash**, **run**.

Message or symptom	Step
`expected ';' before '}'`	
`undefined reference to process_sample`	
`No ST-LINK detected`	
The program stops at the breakpoint, but the variable has the wrong value	

6. Each sentence below is a claim. Write **what you would look at** to prove it: the build console, the download message, the debugger, or the board itself.

   a. "My code compiled."

   b. "The program is inside the chip."

   c. "The loop runs again and again."

   d. "The LED blinks once per second."

Part C - Predict before you run (10 minutes)

7. Read the code. Do **not** run it.

```
1  int count = 0;
2
3  while (count < 3) {
4      count = count + 1;
5  }
```

   a. Fill the table.

After loop pass	`count` is
1	
2	
3	

   b. What is the final value of `count`?

   c. On which line would you put a breakpoint to watch `count` change? Why there?

8. Two programs. Only one number is different.

```
/* Program A */
while (1)
{
    HAL_GPIO_TogglePin(...);
    HAL_Delay(500);
}
```

```
/* Program B */
while (1)
{
    HAL_GPIO_TogglePin(...);
    HAL_Delay(100);
}
```

   a. In 10 seconds, about how many times does the LED toggle in Program A? In Program B?

b. What stays the same in both programs?

c. How would you check your answer with a phone stopwatch?

9. You change `500` to `100`, press **F5**, and the LED blinks exactly as before. Give **three** possible reasons. For each reason, write one quick test.

Possible reason	Quick test
1.	
2.	
3.	

Part D - Ask for help the right way (10 minutes)

10. A student says: *"My code does not work."* Here are the facts: the build and the download succeeded; LD2 should blink every 0.5 s; LD2 stays off; a breakpoint inside `while (1)` is never hit. Write the four lines:

- **Expected:** …
- **Observed:** …
- **Evidence:** …
- **Next test:** …

11. A breakpoint on `HAL_GPIO_TogglePin(...)` is never reached. Number these checks from 1 (do first) to 5 (do last). Then explain your number 1 in one sentence.

Check	Order
Replace the USB cable	
Check that the program reaches `main()`	
Check that the correct project was flashed	
Check whether the program is stopped somewhere else	
Check the loop and `if` conditions	

12. The LED blinks correctly. In the debugger, `toggle_count` still shows `0` after 10 seconds. Can the blinking LED prove that `toggle_count` is correct? **Yes / No.** Give two reasons.

Part E - Your project idea (5 minutes)

13. Choose one device you know: an alarm clock, a kettle, a bike light, a parking sensor, anything. Fill the five boxes with one word or a short phrase each.

Input	Acquisition	Processing	Output	Communication

14. Describe the **smallest version that works** in at most three sentences. Then name **one extra feature** you would add only after the small version works.

Self-check

Mark one:

- ☐ I am ready for Lab 1.
- ☐ I need to review the path from code to LED (Part B).
- ☐ I need help with C loops and variables (Part C).
- ☐ I need help with the VS Code / CubeMX setup.